
September 9, 2026

History

A Crisis in 19th-Century Mathematics

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In 1872, **Karl Weierstrass** broke this assumption — on purpose.

He constructed a function that is **continuous everywhere**, yet **differentiable nowhere**: infinitely jagged, at every single scale, with no tangent line anywhere at all.

Mathematical "Monsters"

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- He had a real problem no one had connected to these ideas.
- He had **computers** — something Cantor, Koch, and Julia never had — letting him see these shapes for the first time.

Definition

What are Fractals?

Definition

A **fractal** is a geometric shape containing detailed structure at arbitrarily small scales, usually having a **fractal dimension** strictly exceeding the **topological dimension**.

Key Characteristics:

- **Scale invariance:** Details at every level of magnification
- **Fractal dimension:** More than the *conventional* dimension
- **Infinite complexity:** Generated by simple, recursive rules
 - Like the infinite complexities of nature

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*"Clouds are not spheres, mountains are not cones,
coastlines are not circles..."*

— **Benoit Mandelbrot**

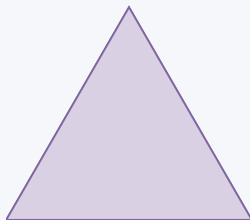
Some Fractals

The Sierpinski Triangle

Let's construct the Sierpinski Triangle:

Construction Process

1. Start with an equilateral triangle



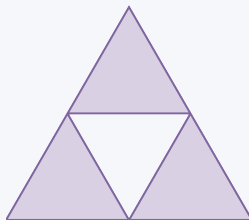
Iteration 0

The Sierpinski Triangle

Let's construct the Sierpinski Triangle:

Construction Process

1. Start with an equilateral triangle
2. Remove the central triangle (triangle connecting midpoints of the sides)



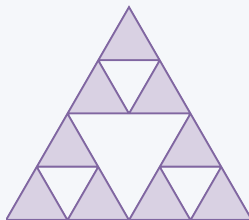
Iteration 1

The Sierpinski Triangle

Let's construct the Sierpinski Triangle:

Construction Process

1. Start with an equilateral triangle
2. Remove the central triangle (triangle connecting midpoints of the sides)
3. Repeat for each triangle



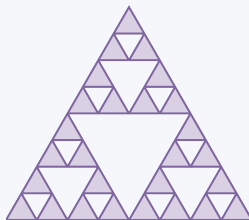
Iteration 2

The Sierpinski Triangle

Let's construct the Sierpinski Triangle:

Construction Process

1. Start with an equilateral triangle
2. Remove the central triangle (triangle connecting midpoints of the sides)
3. Repeat for each triangle
4. After infinite iterations, we get the Sierpinski Triangle



Iteration 3

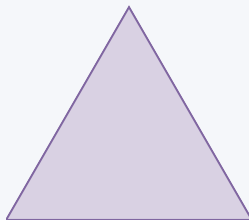
Area & Perimeter

The Sierpinski Triangle: Area

Now let's find the area of triangle(s):

Area Calculation

$$A_0 = \frac{1}{2} \cdot \sin(60^\circ) \cdot a^2 = \frac{\sqrt{3}}{4} a^2$$



Iteration 0

The Sierpinski Triangle: Area

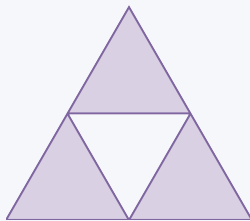
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$$\begin{aligned} A_1 &= A_0 - \frac{1}{2} \cdot \sin(60^\circ) \cdot \left(\frac{a}{2}\right)^2 \\ &= A_0 - \frac{\sqrt{3}}{16} a^2 = A_0 - \frac{1}{4} A_0 \\ &= \frac{3}{4} A_0 \end{aligned}$$

We cut out one fourth of the area at each step.
So, naturally area is reduced by $\frac{3}{4}$ at each step.



Iteration 1

The Sierpinski Triangle: Area

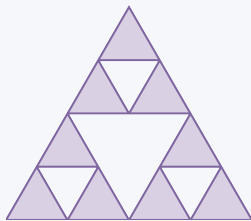
Now let's find the area of triangle(s):

Area Calculation

$$A_0 = \frac{1}{2} \cdot \sin(60^\circ) \cdot a^2 = \frac{\sqrt{3}}{4} a^2$$

$$A_1 = \left(1 - \frac{1}{4}\right) A_0 = \frac{3}{4} A_0$$

$$A_2 = \left(1 - \frac{1}{4}\right) A_1 = \left(\frac{3}{4}\right)^2 A_0$$



Iteration 2

We cut out one fourth of the area at each step.
So, naturally area is reduced by $\frac{3}{4}$ at each step.

The Sierpinski Triangle: Area

Now let's find the area of triangle(s):

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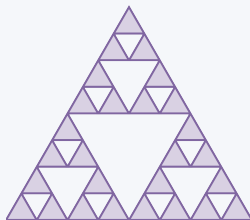
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...

$$A_n = \left(1 - \frac{1}{4}\right)^n A_0 = \left(\frac{3}{4}\right)^n A_0$$

$$\lim_{n \rightarrow \infty} A_n = 0$$



Iteration 3

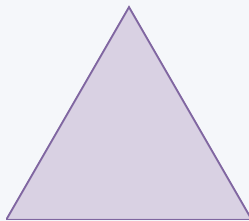
The area of the Sierpinski Triangle is zero even though it is a 2D shape.

The Sierpinski Triangle: Perimeter

Now let's find the perimeter:

Perimeter Calculation

$$P_0 = 3a$$



Iteration 0

The Sierpinski Triangle: Perimeter

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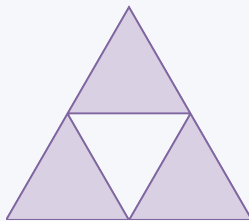
Perimeter Calculation

$$P_0 = 3a$$

$$P_1 = P_0 + 3 \cdot \frac{a}{2} = \frac{9}{2}a = \frac{3}{2}P_0$$

We add half of the previous perimeter at each step. This is because we add the same number of triangles as the last step, but each triangle is half the size of the previous one.

As a result, the perimeter grows by a factor of $\frac{3}{2}$ at each step.



Iteration 1

The Sierpinski Triangle: Perimeter

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Perimeter Calculation

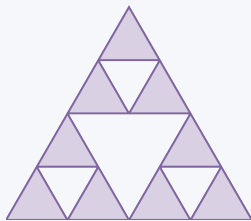
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Iteration 2

The Sierpinski Triangle: Perimeter

Now let's find the perimeter:

Perimeter Calculation

$$P_0 = 3a$$

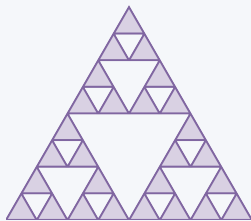
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...

$$P_n = \frac{3}{2}P_{n-1} = \left(\frac{3}{2}\right)^n P_0$$

$$\lim_{n \rightarrow \infty} P_n = \infty$$



Iteration 3

The Sierpinski Triangle has infinite perimeter but zero area.

Handoff to Speaker 2

We've now seen something strange, and precise:

- A **2D shape** (Sierpinski Triangle) with **zero area**.
- A shape built from a rule so simple a child could apply it by hand.
- No matter how far we zoom in, there's always more detail.

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That number is exactly what Hausdorff was reaching for in 1918 — and it's where Speaker 2 picks up the story.

Fractal Dimension

Let's talk about a property of fractals. . .



Fractal Dimension

What does it mean for a shape to have a **non-integer** dimension?

Dimensions We Know

We are all familiar with integer dimensions:



Dimensions We Know

We are all familiar with integer dimensions:



Line



1D



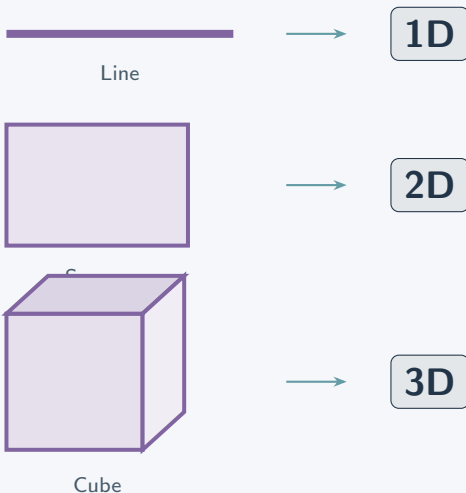
Square



2D

Dimensions We Know

We are all familiar with integer dimensions:



Dimensions are always whole numbers. . . right?

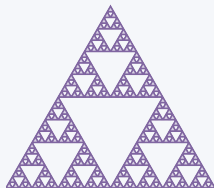
What About This One?

What dimension does this shape
have?

Any guesses?

- It lives in 2D space. . .
- But it has **zero area**
- And **infinite perimeter**

It's not quite 2D, but it's clearly more than
1D. . .



Sierpinski Triangle

What About This One?

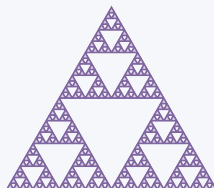
Sierpinski Dimension

The Sierpinski Triangle has a fractal dimension of:

$$D \approx 1.585$$

!!

A **non-integer** dimension — between a line (1D) and a filled surface (2D).



$$D = \log 3 / \log 2 \approx 1.585$$

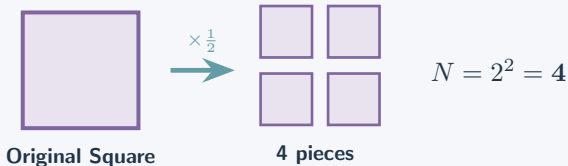
Self-Similarity & Scaling

Scaling a Line ($D = 1$): Scale by $r = \frac{1}{2} \implies 2$ pieces. Scale again by $\frac{1}{2} \implies 2 \times 2 = 4$ pieces.



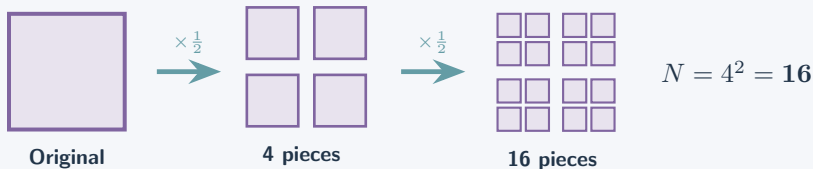
Self-Similarity & Scaling

Scaling a Square ($D = 2$) — Step 1: Scale by $r = \frac{1}{2} \implies$ **4 self-similar pieces** (2×2 grid).



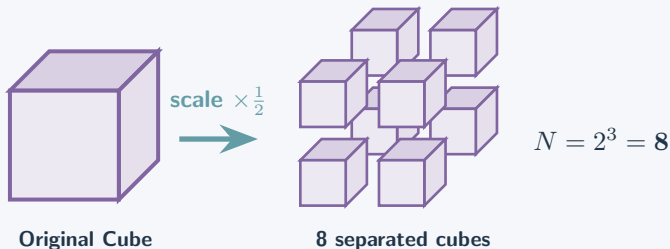
Self-Similarity & Scaling

Scaling a Square ($D = 2$) — Step 2: Scale each piece by $\frac{1}{2}$ again $\implies 4 \times 4 = \mathbf{16}$ self-similar pieces (4×4 grid)!



Self-Similarity & Scaling

Scaling a Cube ($D = 3$): Scale by $r = \frac{1}{2} \implies$
8 self-similar pieces ($2 \times 2 \times 2$ grid in 3D).



Self-Similarity & Scaling

Do You See a Pattern?

Each scaling step multiplies pieces by $(1/r)^D$!

Shape	Scale	1st Step	2nd Step	D
Line	$1/2$	$2 = 2^1$	$4 = 2^2$	1
Square	$1/2$	$4 = 2^2$	$16 = 2^4$	2
Cube	$1/2$	$8 = 2^3$	$(64 = 2^6)$	3

The Pattern

$$\text{Number of pieces} = \left(\frac{1}{r}\right)^D$$



$$D = ?$$

What if the shape is a **fractal**?

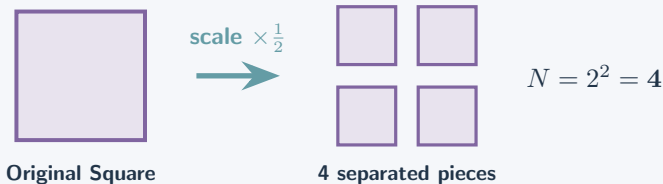
The Self-Similarity Rule

After scaling by factor r , the number of self-similar pieces is:

$$N = \left(\frac{1}{r} \right)^D$$

Square ($D = 2$) with scale factor $r = \frac{1}{2}$:

$$N = \left(\frac{1}{1/2} \right)^2 = 2^2 = 4 \text{ pieces}$$

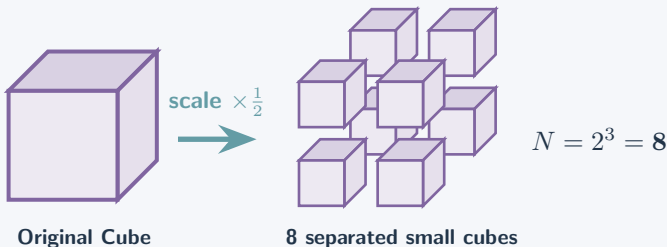


The Self-Similarity Rule

After scaling by factor r , the number of self-similar pieces is:

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Cube ($D = 3$) with scale factor $r = \frac{1}{2}$: $N = \left(\frac{1}{1/2} \right)^3 = 2^3 = 8$ pieces



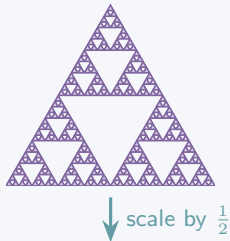
*Can we use this same rule to find the dimension of a **fractal**?*

Dimension of the Sierpinski Triangle

Let's apply the self-similarity rule to the **Sierpinski Triangle**.

Step 1

Scale the Sierpinski Triangle by $\frac{1}{2}$.

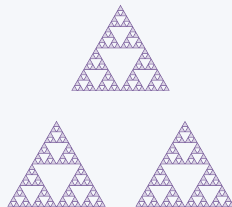


Dimension of the Sierpinski Triangle

Step 2 — Self-Similar Pieces

We get exactly **3** self-similar copies of the original triangle.

Each smaller triangle is a perfect replica, scaled by $r = \frac{1}{2}$.



3 copies

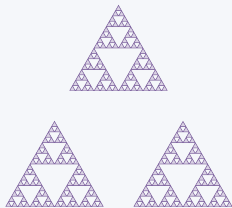
Dimension of the Sierpinski Triangle

Step 3 — The Equation

Using $N = \left(\frac{1}{r}\right)^D$:

$$3 = \left(\frac{1}{1/2}\right)^D$$

$$3 = 2^D$$



3 copies

Dimension of the Sierpinski Triangle

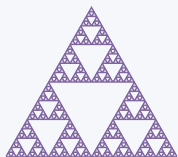
Step 4 — Solving for D

$$2^D = 3$$

$$D \log 2 = \log 3$$

$$D = \frac{\log 3}{\log 2}$$

$$D \approx \mathbf{1.585}$$



$$D \approx 1.585$$

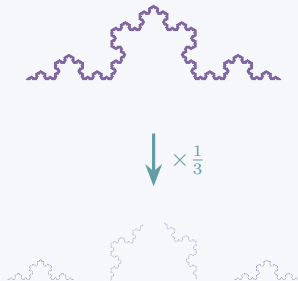
Not an integer! — strictly between 1D (line) and 2D (surface).

Similarly: The Koch Curve

The **Koch Curve** can be divided into **4** self-similar pieces, each scaled by $r = \frac{1}{3}$.

Setup

- Scale factor: $r = \frac{1}{3}$
- Self-similar pieces: $N = 4$



4 copies

Similarly: The Koch Curve

Koch Dimension

$$N = \left(\frac{1}{r}\right)^D$$

$$4 = 3^D$$

$$D = \frac{\log 4}{\log 3}$$

$$D \approx \mathbf{1.262}$$



$$D \approx 1.262$$

Also **non-integer** — between a line (1D) and a surface (2D), but closer to a line than the Sierpinski Triangle.

The General Theory: Box-Counting Dimension

The **Self-Similarity Dimension** works beautifully for self-similar fractals. But what about more complex shapes?

Box-Counting Dimension

The fractal dimension D is defined as:

$$D = \lim_{\varepsilon \rightarrow 0} \frac{\log N(\varepsilon)}{\log(1/\varepsilon)}$$

where $N(\varepsilon)$ is the number of boxes of side length ε needed to cover the fractal.

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where $N(\varepsilon)$ is the number of boxes of side length ε needed to cover the fractal.

Intuition

- Cover the fractal with a grid of boxes of size ε
- Count how many boxes $N(\varepsilon)$ the fractal touches
- Shrink $\varepsilon \rightarrow 0$ and observe how $N(\varepsilon)$ grows
- The **rate of growth** tells us the dimension

Handoff to Speaker 3

We've now uncovered a remarkable property of fractals:

- They have **non-integer dimensions** — genuinely “between” the familiar 1D, 2D, 3D.
- The Sierpinski Triangle lives in dimension $D \approx 1.585$.
- The Koch Curve lives in dimension $D \approx 1.262$.
- The **box-counting dimension** generalises this to *any* shape.

Handoff to Speaker 3

We've now uncovered a remarkable property of fractals:

- They have **non-integer dimensions** — genuinely “between” the familiar 1D, 2D, 3D.
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- The Koch Curve lives in dimension $D \approx 1.262$.
- The **box-counting dimension** generalises this to *any* shape.

We now know *what* fractals are and how to measure their complexity.

But **where do we see them?** And **how are they used** in computer science?

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That's exactly where Speaker 3 takes us — fractals in nature, art, and computation.